*6.40 Determine $i_L(t)$ in the circuit of Fig. P6.40 and plot its waveform for $t \geq 0$.

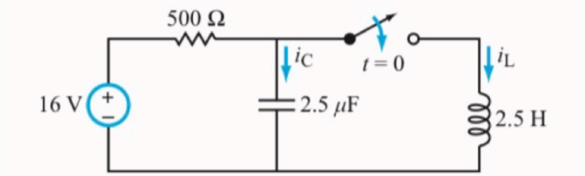
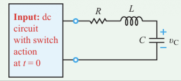
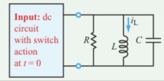
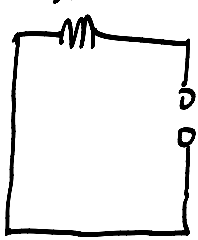


Figure P6.40 Circuit for Problems 6.40 and 6.41.

Table 6-1: Step response of RLC circuits for $t \geq 0$.

Series RLC	Parallel RLC
	
Total Response	**Total Response**
Overdamped ($\alpha > \omega_0$) $\upsilon_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + \upsilon_C(\infty)$ $A_1 = \dfrac{\frac{1}{C} i_C(0) - s_2[\upsilon_C(0) - \upsilon_C(\infty)]}{s_1 - s_2}$ $A_2 = \dfrac{\frac{1}{C} i_C(0) - s_1[\upsilon_C(0) - \upsilon_C(\infty)]}{s_2 - s_1}$	**Overdamped** ($\alpha > \omega_0$) $i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$ $A_1 = \dfrac{\frac{1}{L} \upsilon_L(0) - s_2[i_L(0) - i_L(\infty)]}{s_1 - s_2}$ $A_2 = \dfrac{\frac{1}{L} \upsilon_L(0) - s_1[i_L(0) - i_L(\infty)]}{s_2 - s_1}$
Critically Damped ($\alpha = \omega_0$) $\upsilon_C(t) = (B_1 + B_2 t)e^{-\alpha t} + \upsilon_C(\infty)$ $B_1 = \upsilon_C(0) - \upsilon_C(\infty)$ $B_2 = \frac{1}{C} i_C(0) + \alpha[\upsilon_C(0) - \upsilon_C(\infty)]$	**Critically Damped** ($\alpha = \omega_0$) $i_L(t) = (B_1 + B_2 t)e^{-\alpha t} + i_L(\infty)$ $B_1 = i_L(0) - i_L(\infty)$ $B_2 = \frac{1}{L} \upsilon_L(0) + \alpha[i_L(0) - i_L(\infty)]$
Underdamped ($\alpha < \omega_0$) $\upsilon_C(t) = e^{-\alpha t}(D_1 \cos \omega_d t + D_2 \sin \omega_d t) + \upsilon_C(\infty)$ $D_1 = \upsilon_C(0) - \upsilon_C(\infty)$ $D_2 = \dfrac{\frac{1}{C} i_C(0) + \alpha[\upsilon_C(0) - \upsilon_C(\infty)]}{\omega_d}$	**Underdamped** ($\alpha < \omega_0$) $i_L(t) = e^{-\alpha t}(D_1 \cos \omega_d t + D_2 \sin \omega_d t) + i_L(\infty)$ $D_1 = i_L(0) - i_L(\infty)$ $D_2 = \dfrac{\frac{1}{L} \upsilon_L(0) + \alpha[i_L(0) - i_L(\infty)]}{\omega_d}$
Auxiliary Relations	
$\alpha = \begin{cases} \dfrac{R}{2L} & \text{Series RLC} \\ \dfrac{1}{2RC} & \text{Parallel RLC} \end{cases}$	$\omega_0 = \dfrac{1}{\sqrt{LC}}$
	$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

$$\alpha = \frac{1}{2RC} = \frac{1}{2(500)(2.5\times10^{-6})} = 400\,[s^{-1}]$$



$R = 500\Omega$

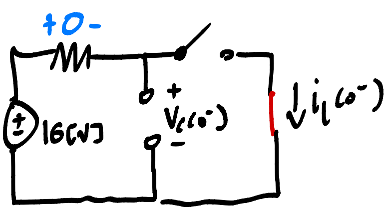
$$\alpha = W_o$$

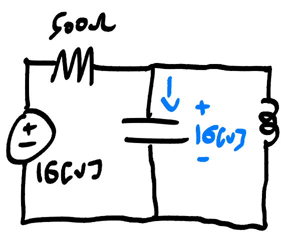
Critically Damped

$$W_o = \frac{1}{\sqrt{LC}} = 400\left[\frac{rad}{s}\right]$$

• $i_L(0^-)$ $t = 0^-$

• $V_L(0^+)$ $t = 0^+$



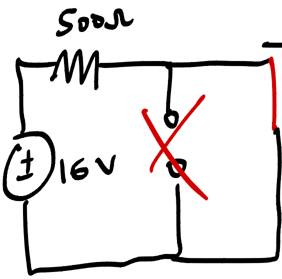


$V_L(0^+) = 16\,[V]$

$V_c(0^-) = 16[V]$

$i_L(0^-) = 0$

$t = \infty \cdot i_L(\infty)$



$$i_L(\infty) = \frac{16\,V}{500\,\Omega} = 0.032\,[A] = 32\,[mA]$$

$$B_1 = i_L(0^-) - i_L(\infty) = 0 - 32[mA] = -32[mA]$$

$$B_2 = \frac{1}{L} v_L(0^+) + \alpha\left(i_L(0^-) - i_L(\infty)\right) = \frac{16[V]}{2.5[H]} + 400\left(-32[mA]\right)$$

$$= -6.4\left[\frac{A}{s}\right]$$

$$i_L(t) = \left(B_1 + B_2 t\right) e^{-\alpha t} + i_L(\infty)$$

$$= \left(-32[mA] - 6.4\left[\frac{A}{s}\right]\right) e^{-400[s^{-1}]t} + 32[mA] \quad t \geq 0$$

$$i_L(t) = x(t)$$

$$x'(t) = i_L'(t)$$

$$v_L(t) = L\, i_L'(t)$$

$$i_L'(t) = \frac{v_L(t)}{L} = \frac{1}{L} v_L(0)$$

*6.29 Choose the value of C in the circuit of Fig. P6.29 so that $v_C(t)$ has a critically damped response for $t \geq 0$. Plot the waveform of $v_C(t)$.

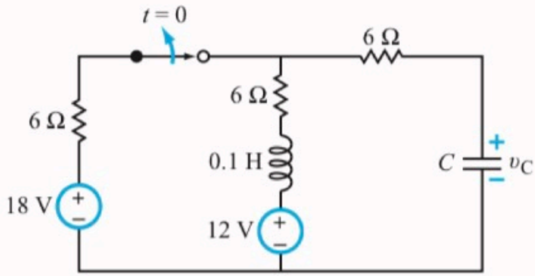
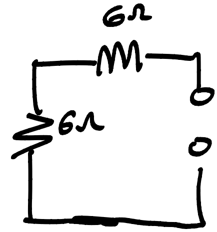


Figure P6.29 Circuit for Problem 6.29.

$\alpha = \frac{R}{2L}$



$R = 12\Omega$

$\alpha = \frac{12}{2(0.1)} = 60\,[s^{-1}]$

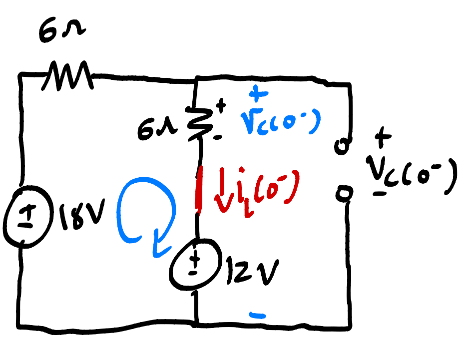
$W_0 = \frac{1}{\sqrt{LC}}$

$\left(\frac{1}{\sqrt{LC}} = 60\,[s^{-1}]\right)^2$

$\Rightarrow \frac{1}{LC} = 60^2$

$C = \frac{1}{L \cdot 60^2} \approx 2.78\,[mF]$

$t = 0^- \cdot V_c(0^-)$



KVL

$-18[V] + 6\Omega \cdot i_L(0^-) + 6\Omega \cdot i_L(0^-) + 12[V] = 0$

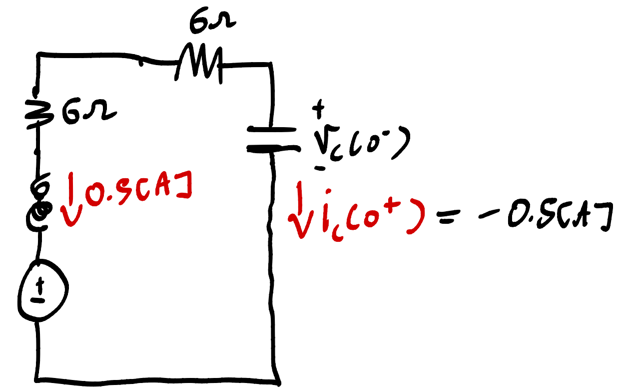
$\Rightarrow 12\Omega\, i_L(0^-) = 6\,[V]$

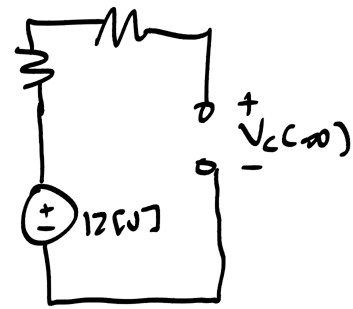
$i_L(0^-) = 0.5\,[A]$

$$V_c(0^-) = 6\Omega \cdot 0.5[A] + 12[V] = 15[V]$$

$t=0^+ \cdot i_c(0^+)$

$t=\infty$





$V_c(\infty) = 12[V]$

$$B_1 = 15 - 12 = 3[V]$$

$$B_2 = \frac{1}{C} i_c(0) + \alpha(B_1) = \frac{1}{2.78\times10^{-3}} \cdot (-0.5) + 60(3)$$

$$-180 + 180 = 0$$

$$\boxed{V_c(t) = 3[V]e^{-60[s^{-1}]t} + 12[V] \quad t \geq 0}$$

*6.46 In the circuit in **Fig. P6.46**, $v_s = 20$ V.

(a) Determine $i_L(t)$ for $t \geq 0$.

(b) If the source is changed to $v_s(t) = e^{-2t}\,u(t)$, can you still use the solution method in part (a) to find $i_L(t)$? If not, why not?

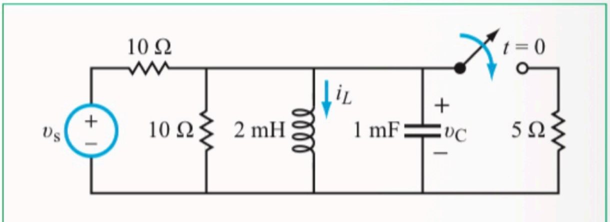
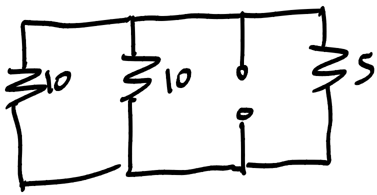


Figure P6.46 Circuit for Problem 6.46.

$$\alpha = \frac{1}{2RC} = \frac{1}{2(2.5)(10^{-3})} = 200\,[s^{-1}]$$

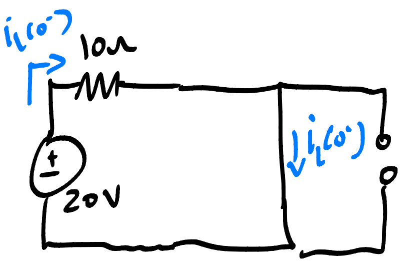
$$\omega_0 = \frac{1}{\sqrt{LC}} = 707.1\,\left[\frac{rad}{s}\right]$$

$$\omega_d = \sqrt{707^2 - 200^2} = 678\,\left[\frac{rad}{s}\right]$$



$R = 2.5\,[\Omega]$

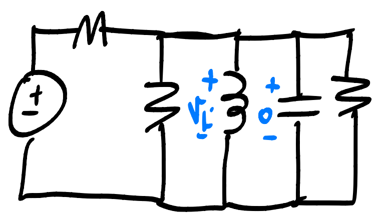
$t = 0^-$: $i_L(0^-)$



$v_C(0^-) = 0$

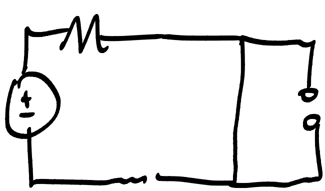
$$i_L(0^-) = \frac{20V}{10\Omega} = 2\,[A]$$

$t = 0^+$



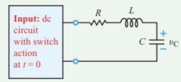
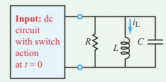
$v_L(0^+) = 0$

$t = \infty$



$i_L(\infty) = 2\,[A]$

Table 6-1: Step response of RLC circuits for $t \geq 0$.

Series RLC	Parallel RLC
	
Total Response	**Total Response**
Overdamped ($\alpha > \omega_0$)	**Overdamped** ($\alpha > \omega_0$)
$v_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_C(\infty)$	$i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$
$A_1 = \dfrac{\frac{1}{C}\, i_C(0) - s_2[v_C(0) - v_C(\infty)]}{s_1 - s_2}$	$A_1 = \dfrac{\frac{1}{L}\, v_L(0) - s_2[i_L(0) - i_L(\infty)]}{s_1 - s_2}$
$A_2 = \dfrac{\frac{1}{C}\, i_C(0) - s_1[v_C(0) - v_C(\infty)]}{s_2 - s_1}$	$A_2 = \dfrac{\frac{1}{L}\, v_L(0) - s_1[i_L(0) - i_L(\infty)]}{s_2 - s_1}$
Critically Damped ($\alpha = \omega_0$)	**Critically Damped** ($\alpha = \omega_0$)
$v_C(t) = (B_1 + B_2 t)e^{-\alpha t} + v_C(\infty)$	$i_L(t) = (B_1 + B_2 t)e^{-\alpha t} + i_L(\infty)$
$B_1 = v_C(0) - v_C(\infty)$	$B_1 = i_L(0) - i_L(\infty)$
$B_2 = \frac{1}{C}\, i_C(0) + \alpha[v_C(0) - v_C(\infty)]$	$B_2 = \frac{1}{L}\, v_L(0) + \alpha[i_L(0) - i_L(\infty)]$
Underdamped ($\alpha < \omega_0$)	**Underdamped** ($\alpha < \omega_0$)
$v_C(t) = e^{-\alpha t}(D_1 \cos\omega_d t + D_2 \sin\omega_d t) + v_C(\infty)$	$i_L(t) = e^{-\alpha t}(D_1 \cos\omega_d t + D_2 \sin\omega_d t) + i_L(\infty)$
$D_1 = v_C(0) - v_C(\infty)$	$D_1 = i_L(0) - i_L(\infty)$
$D_2 = \dfrac{\frac{1}{C}\, i_C(0) + \alpha[v_C(0) - v_C(\infty)]}{\omega_d}$	$D_2 = \dfrac{\frac{1}{L}\, v_L(0) + \alpha[i_L(0) - i_L(\infty)]}{\omega_d}$

Auxiliary Relations

$$\alpha = \begin{cases} \dfrac{R}{2L} & \text{Series RLC} \\[2mm] \dfrac{1}{2RC} & \text{Parallel RLC} \end{cases} \qquad \omega_0 = \frac{1}{\sqrt{LC}} \qquad \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \qquad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$D_1 = 0$$

$$D_2 = \frac{\cancel{\frac{1}{L}\cdot 0}^{\,0} + \cancel{\alpha(0)}^{\,0}}{678} = 0$$

$$i_L(t) = \cancel{e^{-200t}(0 + 0)}^{\,0} + 2[A]$$

$$i_L(t) = 2[A] \quad t \geq 0$$